

Anderson Junior College
Preliminary Examination 2009
H2 Mathematics Paper 1

Answer ALL questions

1. Solve the inequality $\frac{12}{x+1} \leq x$. [2]

Hence solve the inequality $\frac{12}{|x|+1} \leq x$. [2]

2. (i) Find the expansion of $(x-8)^{\frac{1}{3}}$ in ascending powers of x up to and including the term in x^2 . State the set of values of x for which the expansion is valid. [3]
- (ii) Using the expansion found in part (i) and $x = -\frac{17}{8}$, obtain a value for $\sqrt[3]{3}$, giving your result to three decimal places. [2]

3. Mary wanted to buy some chicken, mutton and beef for a barbeque. She visited 3 supermarkets and found that chicken was sold in packs of 2kg, mutton in packs of 0.5kg and beef in packs of 1kg according to the prices stated in the table below.

Super-market	Chicken /pack	Mutton /pack	Beef /pack	Total Amount of money Mary would have to spend
CS	\$7.45	\$4.60	\$8.95	\$89.70
NC	\$7.20	\$4.40	\$8.70	\$86.80
SS	\$7.15	\$4.45	\$8.90	\$87.40

Find the amount of chicken, mutton and beef (in kg) that Mary intended to buy. [3]

4. The first and second terms of an infinite geometric series are a and $(a^2 - a)$ respectively. If all the terms of the series are positive, find the set of values of a for which the series converges. [2]

For this infinite series, it is known that the sum of all the terms after the n th term is equal to the n th term. Find the value of a and hence the sum to infinity of the series. [3]

5. The function f is defined by $f : x \rightarrow e^{x^2-2x} - 2$, $x < 0$.

(i) Find f^{-1} in a similar form. [3]

The function g is defined by $g : x \rightarrow \frac{x-3}{x-1}$, $x > k$, $x \neq 1$.

(ii) Find the least value of k such that the function $f^{-1}g$ exists. [2]

Hence find the range of $f^{-1}g$, giving your answer in exact form. [1]

6. A hexagon $ABCDEF$ has fixed perimeter r cm. Its shape is such that BCD and AEF are equilateral triangles, while $ABDE$ is a rectangle. If the length of BC is s cm, show that the area of the hexagon is given by $\left(\frac{\sqrt{3}}{2} - 2\right)s^2 + \frac{rs}{2}$ cm². [2]

Hence find, in terms of r , the maximum area of the hexagon, leaving your answer in exact form. [3]

7. (a) Find $\frac{d}{dx}\left[\cos^{-1}\left(\frac{1}{x^2}\right)\right]$ and simplify your answer. [2]

- (b) The curve C has parametric equations $x = \ln(\sin t)$, $y = \cot t$, where $0 < t < \pi$.

Show that $\frac{dy}{dx} = -\frac{2}{\sin 2t}$. [2]

P is the point on the curve C where the normal is parallel to the line $2y = x - 2$.

Find the equation of the normal at P . [2]

This normal intersects the x -axis at the point Q . R is a point on the curve C such that $PQ = RQ$. Find the equation of the normal to the curve C at the point R . [2]

8. Find

(i) $\int e^{2x} \tan^{-1}(e^{-2x}) dx$ [3]

(ii) $\int \frac{x}{\sqrt{1-4x-2x^2}} dx$. [3]

9. The curve C has equation $y = \frac{x^2 - ax + b}{x - 2}$ where a, b are constants and $b \neq 0$.

- (i) Find $\frac{dy}{dx}$ and hence show that if the curve C has 2 stationary points, then $b > 2a - 4$. [3]

- (ii) Given that $-1 < b < 0$, sketch the curve C for $a = 5$, showing clearly all its asymptotes. [2]

Hence find the largest possible value of k such that the equation

$2x^2 - 10x - 1 = (mx - 3)(2x - 4)$ has exactly 2 real roots for all $m < k$. [2]

10. The complex number z satisfies both the conditions

$$5 \leq |z - 3 - 4i| \leq 10 \text{ and } \arg\left(\frac{3 + 4i}{z - 6 - 8i}\right) = 0.$$

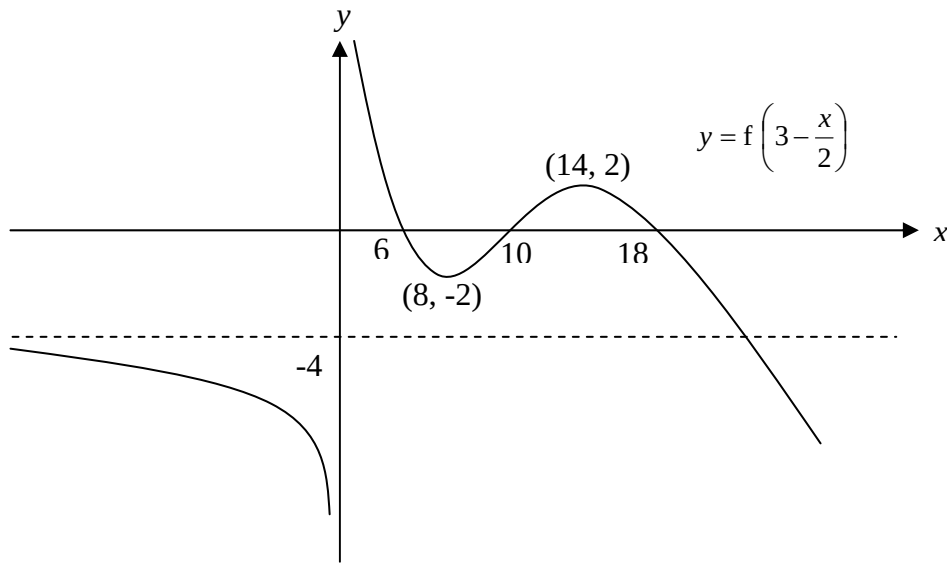
Sketch the locus of the points representing the complex number z on an Argand diagram. [4]

Hence find

- (i) the maximum value of $|z - 6|$, giving your answer in exact form, [2]

- (ii) the range of values of $\arg(z - 6)$. [2]

11. The graph of $y = f\left(3 - \frac{x}{2}\right)$ has 2 stationary points at $(8, -2)$ and $(14, 2)$ and intersects the x -axis at $x = 6$, $x = 10$ and $x = 18$ as shown in the diagram below.



Sketch, on separate clearly labeled diagrams, the graphs of

(a) $y = \frac{1}{f\left(3 - \frac{x}{2}\right)}$ [3]

(b) $y^2 = -f\left(3 - \frac{x}{2}\right)$ [3]

(c) $y = f(x)$ [3]

Show clearly the asymptotes, stationary points and points of intersection with the axes in your diagrams.

12. A sequence of real positive numbers $u_1, u_2, u_3, \dots$ satisfies the recurrence relation

$$u_{n+1} = u_n + 4 + 4\sqrt{u_n + 1}, \quad n \in \mathbb{N}^+ \quad \text{and} \quad u_1 = 15.$$

(i) Prove by induction that $u_n = 4(n+1)^2 - 1$ for all $n \in \mathbb{N}^+$. [4]

(ii) Express $\frac{1}{u_n}$ in the form $\frac{A}{2n+1} + \frac{B}{2n+3}$, where A and B are constants to be determined. [2]

(iii) Hence find $\sum_{n=1}^N \frac{1}{u_n}$ in terms of N . [2]

(iv) Using your answer in part (iii), find $\sum_{n=1}^N \frac{1}{4n^2 - 1}$. [3]

13. The points P and Q have position vectors $\mathbf{i} - \mathbf{j}$ and $3\mathbf{i} + 13\mathbf{j} + 6\mathbf{k}$ respectively. The plane π_1 contains the point P and the line $\frac{x}{2} = -1 - z, y = 0$.

- (i) Find a vector equation of the plane π_1 in scalar product form. [3]
(ii) Find the position vector of the foot of the perpendicular from Q to π_1 . [2]

The line l_1 passes through the points P and Q .

- (iii) The line l_2 is the reflection of the line l_1 about the plane π_1 . Find a vector equation of l_2 . [3]

The plane π_2 has the equation $\mathbf{r} \cdot \begin{pmatrix} a \\ 6 \\ 4 \end{pmatrix} = b$.

Find the values of a and b such that

- (iv) π_1 and π_2 are parallel and at a distance of $\sqrt{224}$ apart. [3]
(v) π_1 and π_2 are intersecting. [1]

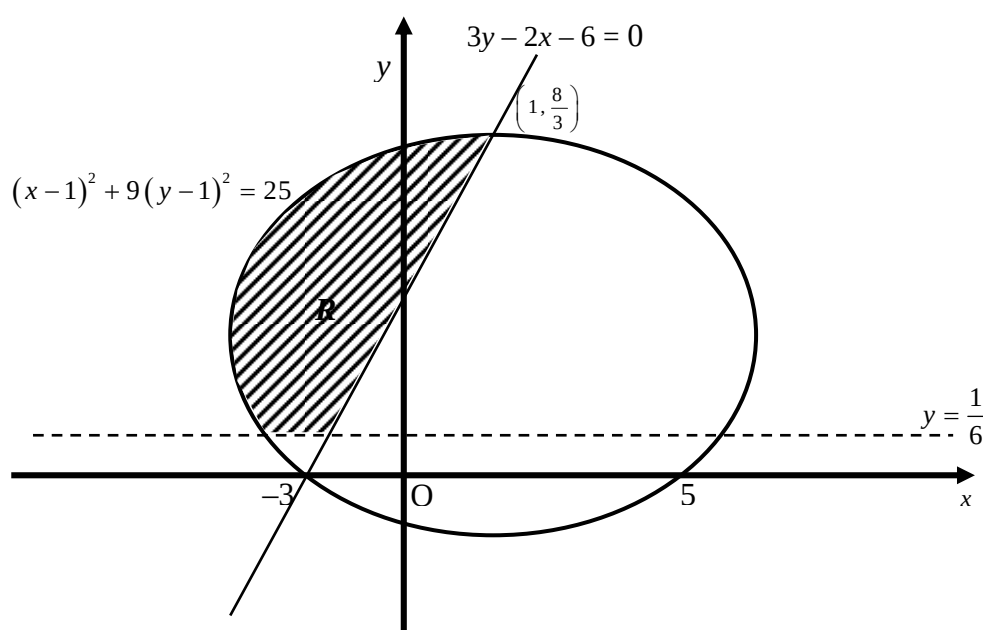
14. (a) (i) By using the substitution $3(y - 1) = 5\sin \theta$, prove that

$$\int_{\frac{1}{6}}^{\frac{8}{3}} \sqrt{25 - 9(y - 1)^2} \, dy = 25 \left(\frac{\pi}{9} + \frac{\sqrt{3}}{24} \right). \quad [4]$$

- (ii) The curve C has equation $(x - 1)^2 + 9(y - 1)^2 = 25$.

The region R is bounded by the curve C and the lines $3y - 2x - 6 = 0$ and

$y = \frac{1}{6}$ as shown in the diagram below. Find the exact area of the region R . [3]



- (b) The region S is the region inside the curve C such that $3y - 2x - 6 \geq 0$. Find the volume of the solid formed when the region S is rotated about the x -axis through four right angles. [4]

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